Scrapy cram sheet

scrapy@2.19.0 · Python 3.11+ · 2026-11-02

If you only read one thing: non-2xx responses and repeated URLs never reach your callback. Check HTTPERROR_ALLOWED_CODES and dont_filter before debugging selectors.

Legend: ★ most used · ⚠ gotcha

DAILY COMMANDS

Command	Does
`scrapy crawl NAME -O out.jsonl` ★	run spider, overwrite output
`scrapy shell URL` ★	console with `response` loaded
`scrapy parse URL --spider NAME -c cb`	run one callback on one URL
`scrapy runspider file.py`	run a spider without a project
`scrapy genspider NAME DOMAIN`	new spider from template
`scrapy fetch --curl 'curl ...'`	fetch a request given as curl
`scrapy genrequest 'curl ...'`	print Request code for curl
`scrapy settings --get NAME`	print a resolved setting

GLOBAL FLAGS

Flag	Does
`-s NAME=VALUE` ★	override a setting (priority 40)
`-a NAME=VALUE`	spider argument, becomes attribute
`-o F` / `-O F`	append / overwrite feed
`-L INFO`	log level (default `DEBUG`)
`--pdb`	debugger on logged exceptions

MINIMAL SPIDER

```
class QSpider(scrapy.Spider):
  name = "q"
  start_urls = [
   "https://quotes.toscrape.com"]

  def parse(self, resp):
    for q in resp.css(".quote"):
      yield {"t": q.css(
        ".text::text").get()}
    yield from resp.follow_all(
      css="li.next a",
      callback=self.parse)
```

- Yield dict → pipelines/feeds; yield `Request` → scheduler
- Custom first requests: `async def start(self)` (2.13+)

SELECTORS (ON `RESPONSE`)

- `.css("h1::text").get()` → first str or `None`
- `.css("a::attr(href)").getall()` → list
- `.xpath("//a/@href").get()` → XPath form
- `.follow(href, callback=cb)` → `Request`, relative URL ok

SETTINGS THAT MATTER

Built-in default → `startproject` template value:

- `ROBOTSTXT_OBEY` False → True
- `CONCURRENT_REQUESTS_PER_DOMAIN` 8 → 1
- `DOWNLOAD_DELAY` 0 → 1
- `RETRY_TIMES` 2 (3 attempts), `DOWNLOAD_TIMEOUT` 180 s
- Priority: default 0 < command 10 < addon 15 < project 20 < spider 30 < cmdline 40

GOTCHAS

- ⚠ 404/500 dropped by `HttpErrorMiddleware`. Use `meta={"handle_httpstatus_list":[404]}`.
- ⚠ Duplicate URLs silently filtered. `dont_filter=True` to revisit.
- ⚠ `allowed_domains` wants `"example.com"`, not a URL.
- ⚠ `-o x.json` twice = two arrays, invalid JSON. Use `-O` or `.jsonl`.
- ⚠ `runspider` outside a project uses built-in defaults: no robots.txt, 8 req/domain.

FIND IT IN THE REPO

- `scrapy/core/engine.py` crawl loop
- `scrapy/core/scheduler.py` queue, dupefilter
- `scrapy/settings/` defaults in `default_settings.py`
- `scrapy/commands/` one file per command
- `tests/mockserver/` server the tests crawl

RUN FROM A SCRIPT

```
from scrapy.crawler import (
  AsyncCrawlerProcess)
p = AsyncCrawlerProcess({
  "FEEDS": {"out.json":
    {"format": "json"}}})
p.crawl(QSpider)
p.start()  # blocks until done
```

ENABLE A PIPELINE

```
ITEM_PIPELINES = {
  "proj.pipelines.Clean": 300}
```

- Lower number runs first; keep to 0–1000
- Same dict shape for `DOWNLOADER_MIDDLEWARES`, `SPIDER_MIDDLEWARES`

CONTRIBUTE

```
pre-commit run --all-files
tox -e mypy && tox -e pylint
tox -e py -- tests/test_foo.py
```